Chap. 3 Strain

Definition of

$\underset{\sim}{\varepsilon}, \underset{\sim}{\gamma}$

$\varepsilon_{x'}, \varepsilon_{y'}, \gamma_{x'y'}$

$\varepsilon_1, \varepsilon_2, \gamma_{max}, m.h$

3.1 Introduction

- In Chapter 1, our concern was with the stresses within a body subject to a system of external forces.

- We now turn to the deformations caused by these forces and to a measure of deformational intensity called strain.

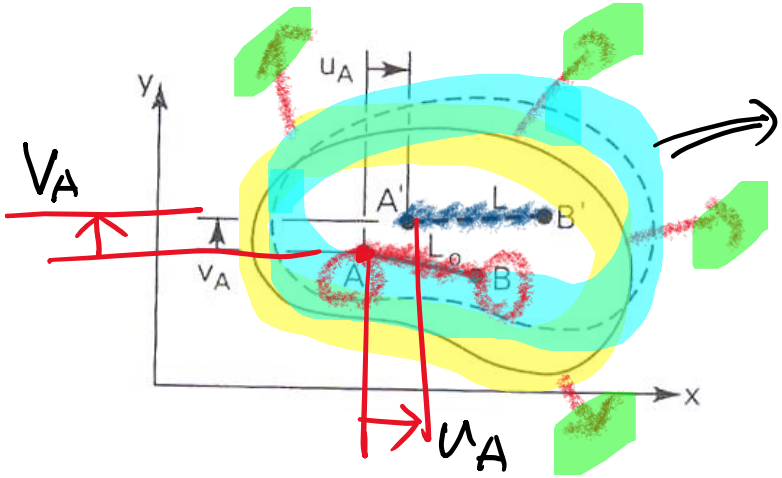
We need deformation

from external loadings!!

Definition of strain

⇒ Displacement!!

3.2 Deformation



Change of position of material
Can be generated by ext. loadings
defined as displacement

Let us consider a body subjected to external loading that causes it to take up the position pictured by the dashed lines.

The displacements of any two points such as A and B are simply AA' & BB', respectively. → Deformation (straining) + rigid body motion (translation & rotation).

- Position change — Rigid body motion
  - deformation

displacement

- To describe the magnitude & direction of the displacements, points within the body are located with respect to an appropriate coordinate reference as, the xyz system.

- In the two-dimensional case, the components of displacement of point A to A' can be represented by u & v in x, y coordinate.

- In the three-dimensional case, u, v, w.

- The displacement at every point within the body constitutes the displacement field, u = u(x, y, z), v = v(x, y, z), w = w(x, y, z).

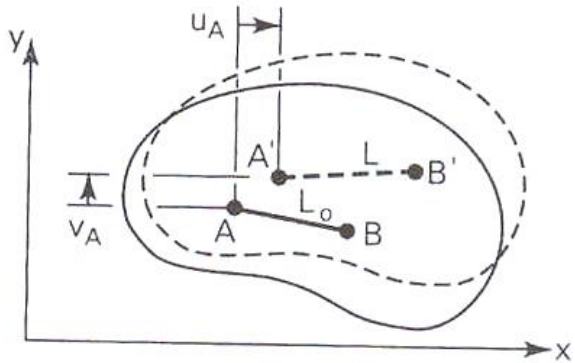
- Only small displacements are considered in this text.

u, v, w ; $\underset{\sim}{u} = u\underset{\sim}{i} + v\underset{\sim}{j} + w\underset{\sim}{k}$ (Rigid body motion, deformation

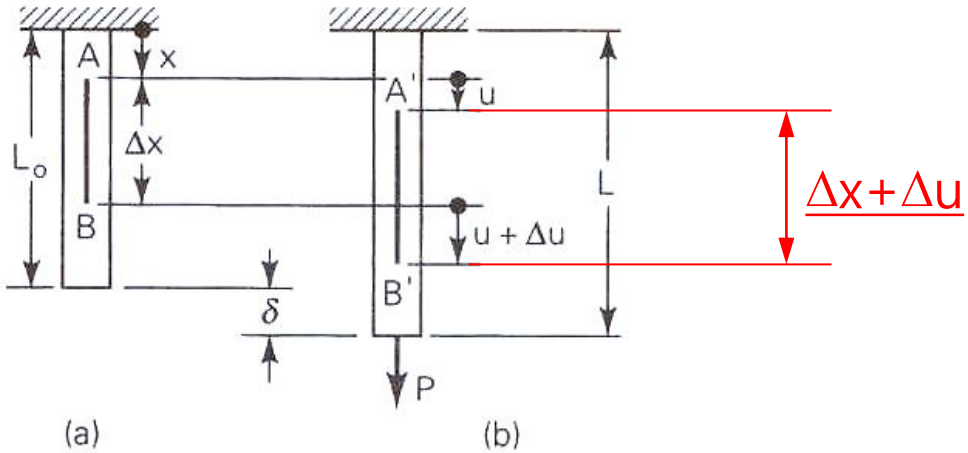
$u(x,y,z)$ $v(x,y,z)$ $w(x,y,z)$

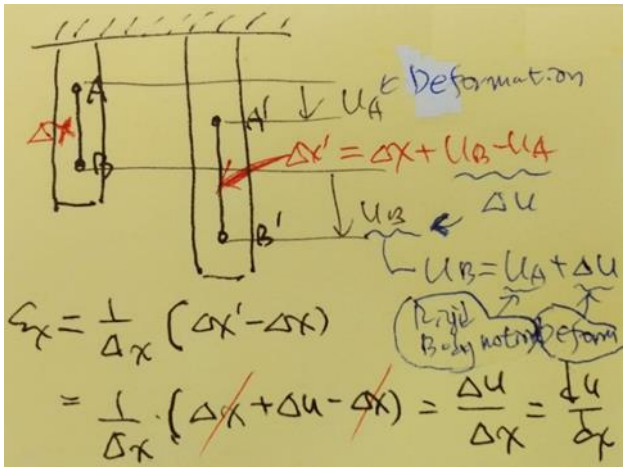
Superposition

▼ Read Page 52. “Superposition”



3.3 Strain defined





For purpose of defining normal strain, refer to Fig.(a)&(b), where line AB of an axially loaded member has suffered deformation to become A'B'.

- The length of AB is Δx.
- The point, A and B have each been displaced; A an amount u, and B, u + Δu.
- Then the length Δx has been increased by Δu.

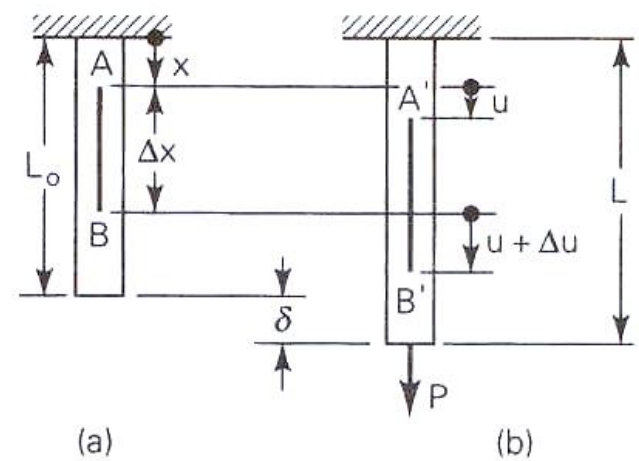
Normal strain, the unit change in length, is defined as

$$\varepsilon = \lim_{\Delta x \to 0} \frac{\Delta x + \Delta u - \Delta x}{\Delta x} = \lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} = \frac{du}{dx} \quad \Leftarrow \quad \varepsilon = \frac{L_f - L_0}{L_0}$$

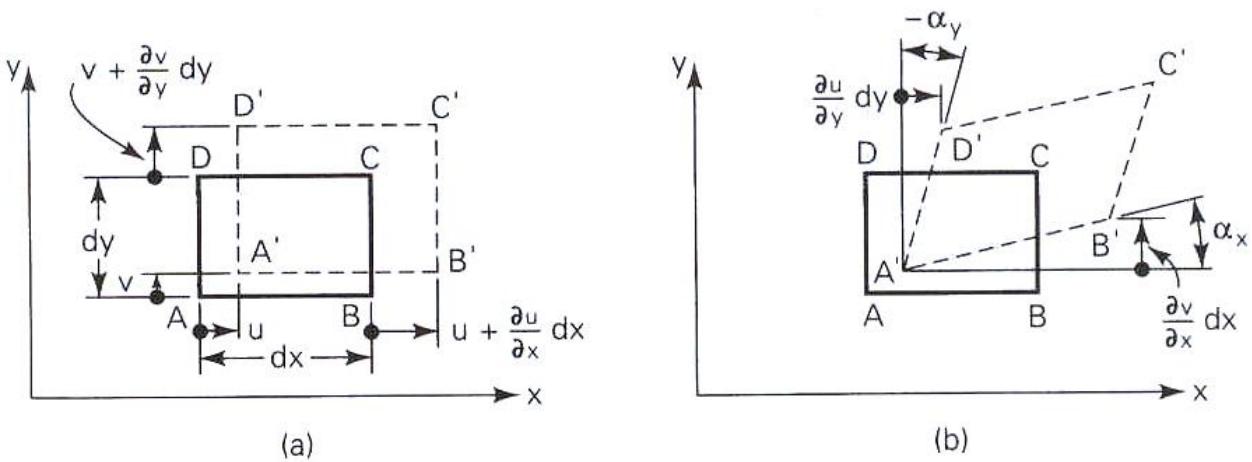
represents the strain at a point, the point to which Δx shrinks.

If the deformation is distributed uniformly over the original length, the normal strain may be written

$$\varepsilon = \frac{\int du}{\int dx} = \frac{L - L_0}{L_0} = \frac{\delta}{L_0}$$



We now investigate the case of two-dimensional or plane strain.



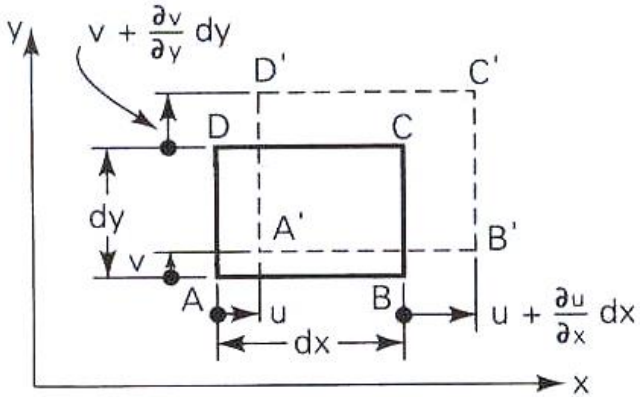
Consider an element with dimensions dx, dy and of unit thickness.

The total deformation may be regarded as a change in length experienced by the sides & a relative rotation without accompanying changes of length.

Two normal or longitudinal strains are

$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_y = \frac{\partial v}{\partial y}$$

A positive sign applied to elongation; a negative sign, to contraction.



(a)

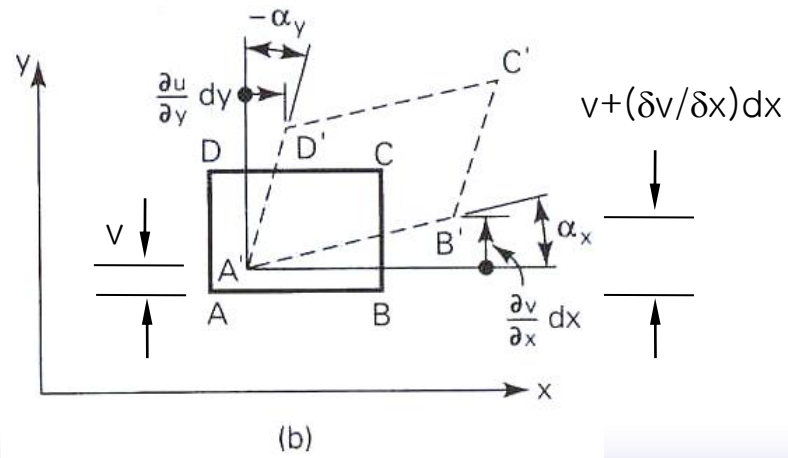
- Now consider the change experienced by right angle DAB.
  We assume very small angle α_x between AB & A'B' then $\alpha_x \approx \tan\alpha_x$, and AB ≈ A'B', $\alpha_x \approx \partial v/\partial x$.

  where counterclockwise rotation is defined as positive.

  Also $-\alpha_y \approx \partial u/\partial y$.

  The angular change between lines in the x & y directions is defined as the shearing strain and denoted by γ_{xy};

$$\gamma_{xy} = \alpha_x - \alpha_y = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$



(b)

The shear strain is positive when the right angle between two positive (or negative) axes decreases. => positive γ_{xy}

$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_y = \frac{\partial v}{\partial y}, \quad \varepsilon_z = \frac{\partial w}{\partial z}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}, \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}, \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

The angular change is not different if it is said to occur between the x and y directions or between the y and x directions.

also $\gamma_{xy} = \gamma_{yx}, \gamma_{yz} = \gamma_{zy}, \gamma_{xz} = \gamma_{zx}$; "Symmetry"

Then the tensor notation of ε_{ij} is

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \quad i, j = 1,2,3$$

$$\left.\begin{matrix} \varepsilon_{xx}, & \gamma_{xy} \\ \varepsilon_{yy}, & \gamma_{yz} \\ \varepsilon_{zz}, & \gamma_{zx} \end{matrix}\right\} \Rightarrow \varepsilon_{ij}$$

The longitudinal strains are obtained when i = j ; the shearing strains are found when i ≠ j.

$$\varepsilon_{xx} = \varepsilon_x = \frac{1}{2}\left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial x}\right) = \frac{\partial u}{\partial x}$$

Then,

$$\varepsilon_{xy} = \frac{1}{2}\gamma_{xy}, \varepsilon_{yz} = \frac{1}{2}\gamma_{yz}, \varepsilon_{xz} = \frac{1}{2}\gamma_{xz}$$

$$\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \rightarrow \gamma_{xy}$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}, \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}, \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

Just as the states of stress at a point is described by a nine-term array, the strain may be presented nine-term composing the symmetric strain tensor ($\varepsilon_{ij} = \varepsilon_{ji}$)

ε_{xy}

$$[\varepsilon_{ij}] = \begin{bmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z \end{bmatrix} \quad <=> \quad [\tau_{ij}] = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

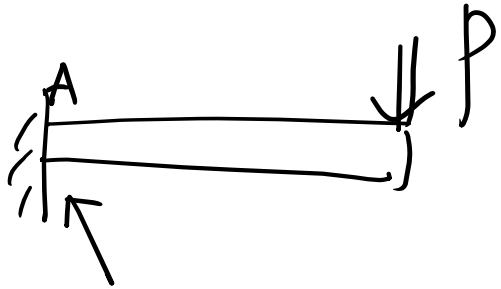
ε_{zx} ε_{zy}

3.4 Equations of compatibility

The concept of compatibility has both mathematical & physical significance.

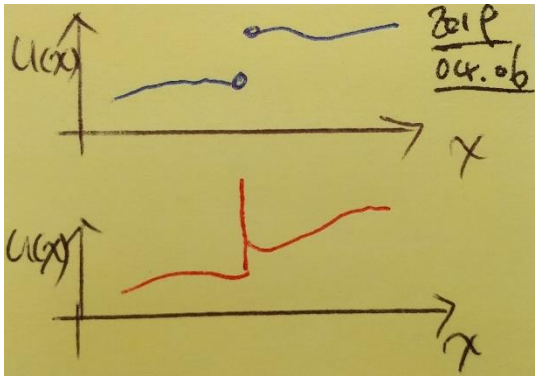
Strain ← displacements
"$\frac{du}{dx}$"

- Mathematical view point :
  It asserts that the displacement u, v, & w match the geometrical conditions and are single-valued and continuous functions of position with which the strain components are associated.



Geometric Compatibility $\left\{ \begin{array}{l} u_A = 0 \\ u_A' = 0 \end{array} \right\}$

slope



No defects

- Physical view point :
  This means the body must be pieced together ; no voids are created in the deformed body.

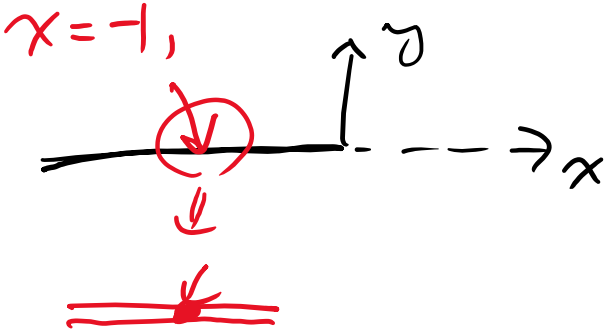
No cracks

=> The kinematics relations between strain & displacement connect six components of strain to only three components of displacement.

▼ Therefore, we can't arbitrarily specify all the strains as functions of x, y, & z.

Must exist relations b.t.w strains !!

based on their definition using displacement !!



As the strains are evidently not independent of one another, in what way are they related?

▼ In two dimensional strain, differentiation of ε_x twice with respect to y, ε_y twice with respect to x, and γ_{xy} with respect to x & y, results in

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} = \frac{\partial^3 u}{\partial x \partial y^2}, \quad \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^3 v}{\partial x^2 \partial y}, \quad \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} = \frac{\partial^3 u}{\partial x \partial y^2} + \frac{\partial^3 v}{\partial x^2 \partial y}$$

=> $$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y}$$

$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_y = \frac{\partial v}{\partial y},$$

$$\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}, \quad \gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}, \quad \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

- This is the condition of compatibility of the two-dimensional problem expressed in terms of strain.

The three-dimensional equations of compatibility are obtained in a like manner :

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \qquad 2\frac{\partial^2 \varepsilon_x}{\partial y \partial z} = \frac{\partial}{\partial x}\left(-\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right)$$

$$\frac{\partial^2 \varepsilon_y}{\partial z^2} + \frac{\partial^2 \varepsilon_z}{\partial y^2} = \frac{\partial^2 \gamma_{yz}}{\partial y \partial z} \qquad 2\frac{\partial^2 \varepsilon_y}{\partial z \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial \gamma_{yz}}{\partial x} - \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right)$$

$$\frac{\partial^2 \varepsilon_z}{\partial x^2} + \frac{\partial^2 \varepsilon_x}{\partial z^2} = \frac{\partial^2 \gamma_{xz}}{\partial z \partial x} \qquad 2\frac{\partial^2 \varepsilon_z}{\partial x \partial y} = \frac{\partial}{\partial z}\left(\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} - \frac{\partial \gamma_{xy}}{\partial z} \right)$$

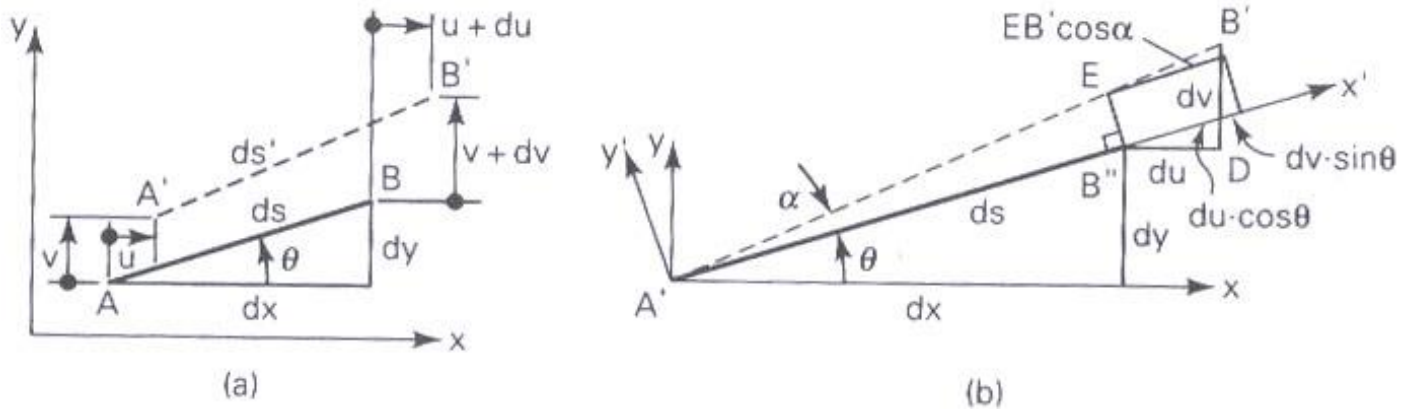
6-compatibility eqns!

These equations were first derived by Saint-Venant in 1860.

- The deformed system will in general, be impossible to arrange in such a way as to compose a continuous body unless the components of strain satisfy the equations of compatibility.

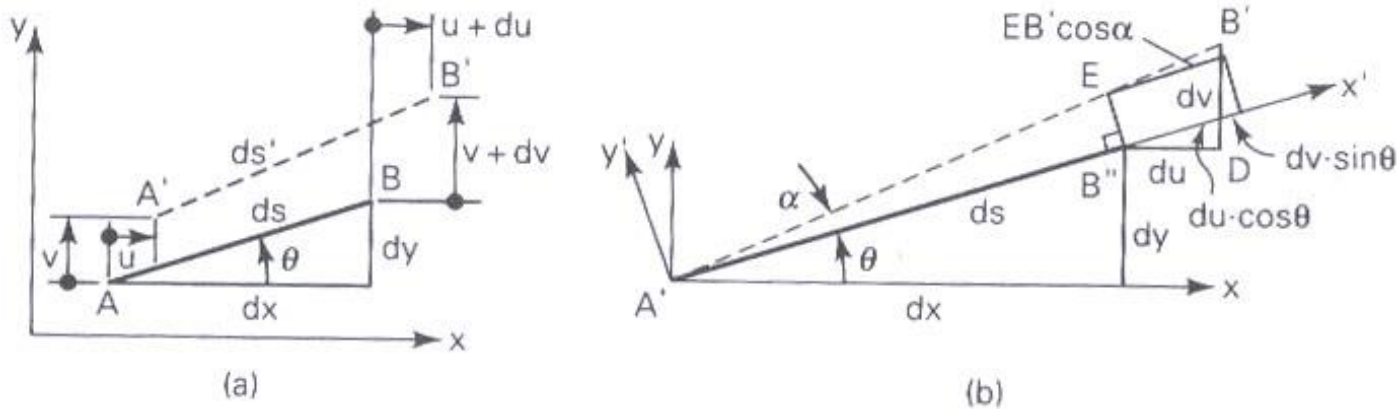
▼ Malvern P.184

3.5 Two-dimensional strain at a point



- Consider a small linear element AB of length ds is an unstrained body. The projections of the element on the coordinate axes are dx & dy

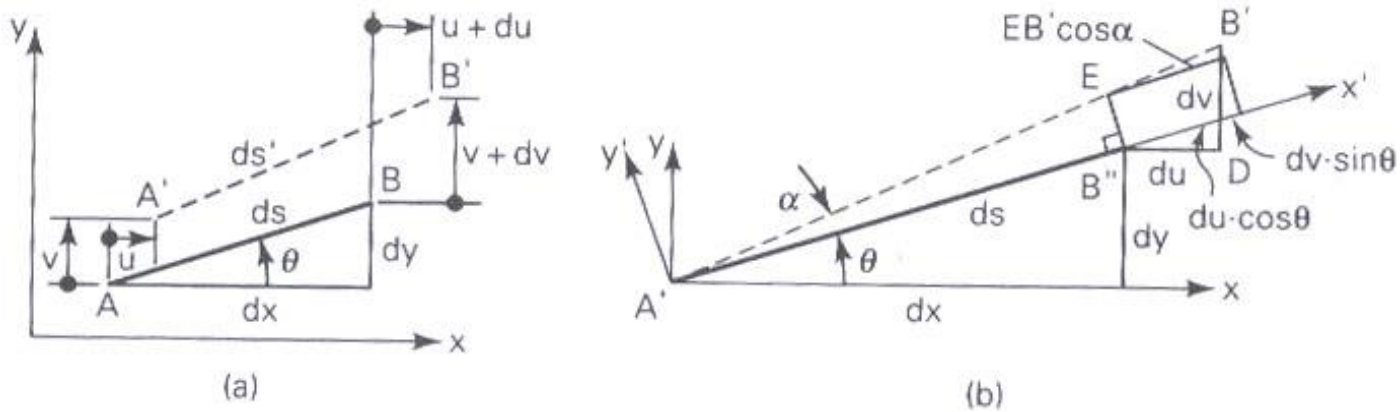
- After straining, AB is displaced to position A'B' and is now ds' long.



The x and y displacement is expressed by a truncated Taylor's Expansion (with two variables) as follows:

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy, \quad dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy \qquad \text{(a)}$$

- See Fig. (b)



- AB has been translated so that A coincides with A' ; it is now in the position A'B". Here, B"D = du & DB' = dv are the components of displacement

Transformation of Two-Dimensional strain

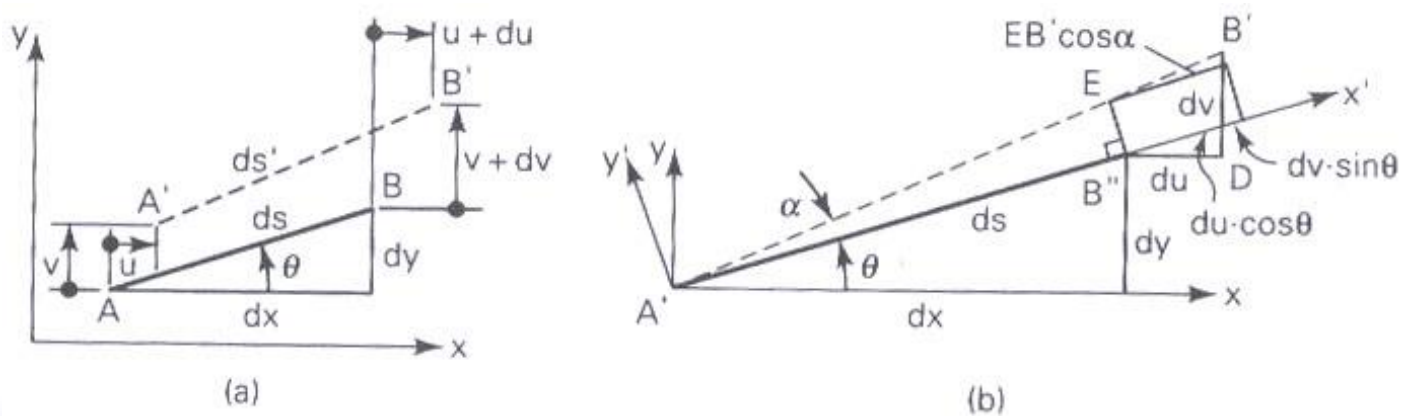
We now choose a new coordinate system x'y' as shown in Fig.(b)

Then,

$$EB'\cos\alpha \approx EB'(small\ angle) = du\cos\theta + dv\sin\theta$$

$$\varepsilon_{x'} = \frac{EB'}{ds}$$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy, \quad dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$



Then,

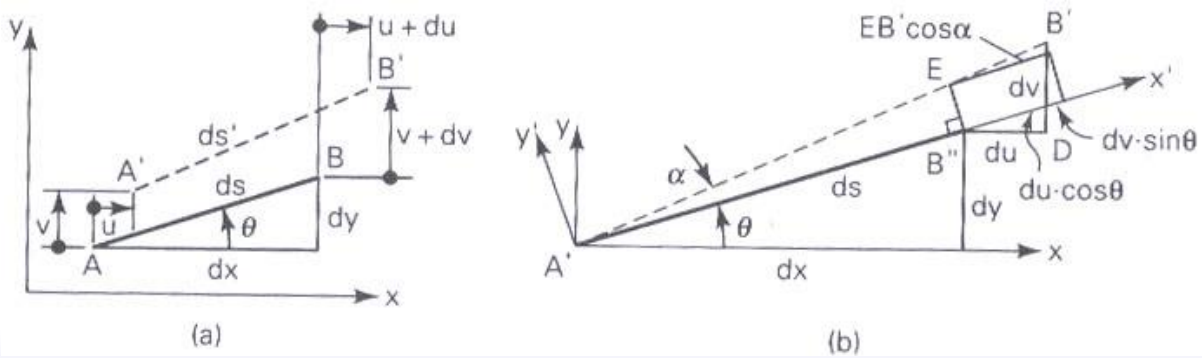
$$\varepsilon_{x'} = \left(\frac{\partial u}{\partial x}\frac{dx}{ds} + \frac{\partial u}{\partial y}\frac{dy}{ds}\right)\cos\theta + \left(\frac{\partial v}{\partial x}\frac{dx}{ds} + \frac{\partial v}{\partial y}\frac{dy}{ds}\right)\sin\theta$$

Where

$$\frac{dx}{ds} = \cos\theta, \frac{dy}{ds} = \sin\theta$$

$$\boxed{\sigma_{x'} = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta}$$

$$\varepsilon_{x'} = \varepsilon_x \cos^2\theta + \varepsilon_y \sin^2\theta + \gamma_{xy}\sin\theta\cos\theta$$



- This represents the transformation equation for the x-directed normal strain.

$$\varepsilon_{x'} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta \quad <=> \quad \boxed{\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta}$$

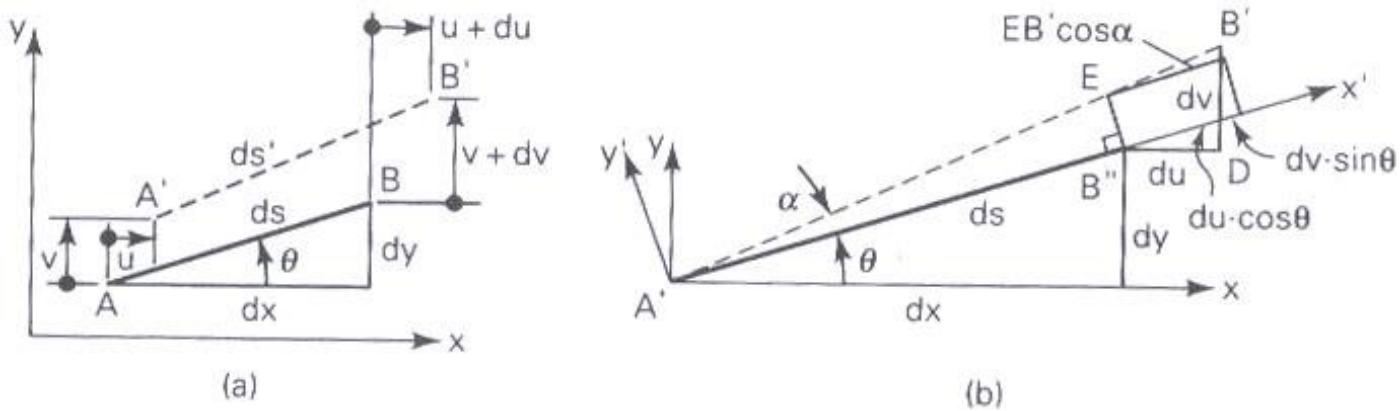
ε_x ε_y $\frac{\gamma_{xy}}{2}$

- The normal strain $\varepsilon_{y'}$ is determined by replacing θ by $\theta + \pi/2$.

Analogous !!

To derive an expression for the shearing strain $\gamma_{x'y'}$,
we first determine the angle α through which AB (the x' axis) is rotated.

$$\tan\alpha = \frac{B''E}{ds}, \quad B''E = \left(dv\cos\theta - du\sin\theta - EB'\sin\alpha\right)$$



$$\boxed{\tan\alpha = \frac{B''E}{ds}, \quad B''E = \left(dv\cos\theta - du\sin\theta - EB'\sin\alpha\right)}$$

By letting

$$\boxed{\varepsilon_{x'} = \frac{EB'}{ds}}$$

(very small X very small)

$$\sin\alpha \approx \tan\alpha \approx \alpha \quad \rightarrow EB'\sin\alpha = \varepsilon_{x'} ds\alpha \approx 0 \quad Since\ \alpha \approx 0$$

Substituting Eqs.(a) and (2.3) into B"E, α=B"E/ds may be written as follows:

$$\alpha = -(\varepsilon_x - \varepsilon_y)\sin\theta\cos\theta + \frac{\partial v}{\partial x}\cos^2\theta - \frac{\partial u}{\partial y}\sin^2\theta$$

$$\boxed{\alpha = B''E/ds = \left(dv\cos\theta - du\sin\theta\right)/ds}$$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy, \quad dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy$$

$$\varepsilon_x = \frac{\partial u}{\partial x},$$

$$\frac{dx}{ds} = \cos\theta, \frac{dy}{ds} = \sin\theta$$

The angular displacement of y' is readily derived by replacing θ by θ + π/2.

$$\alpha\Big|_{\theta=\theta+\frac{\pi}{2}} = (\varepsilon_x - \varepsilon_y)\sin\theta\cos\theta + \frac{\partial v}{\partial x}\sin^2\theta - \frac{\partial u}{\partial y}\cos^2\theta \quad <=> \quad \boxed{\alpha = -(\varepsilon_x - \varepsilon_y)\sin\theta\cos\theta + \frac{\partial v}{\partial x}\cos^2\theta - \frac{\partial u}{\partial y}\sin^2\theta}$$

Then,

$$\gamma_{x'y'} = \alpha_\theta - \alpha_{\theta+\frac{\pi}{2}} = -2(\varepsilon_x - \varepsilon_y)\sin\theta\cos\theta + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)(\cos^2\theta - \sin^2\theta)$$

▼ Use of trigonometric identities

$$\gamma_{x'y'} = -(\varepsilon_x - \varepsilon_y)\sin 2\theta + \gamma_{xy}\cos 2\theta$$

$$<=> \quad \boxed{\tau_{x'y'} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau_{xy}\cos 2\theta}$$

$$\frac{\gamma_{x'y'}}{\cancel{2}} = -\frac{1}{\cancel{2}}(\varepsilon_x - \varepsilon_y)\sin 2\theta + \frac{\gamma_{xy}}{\cancel{2}}\cos 2\theta$$

It is observed that transformation expressions for stress are converted into strain relationships by replacing

σ with ε & τ with γ/2

▼ These substitutions can be made in all the analogous relations.

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \pm \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2}$$

$$\tan 2\theta_p = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y}$$

$$\gamma_{max} = \pm 2\sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \left(\frac{\gamma_{xy}}{2}\right)^2} = \pm(\varepsilon_1 - \varepsilon_2)$$

<=>

$\sigma_{ave} \pm R$

$$\sigma_{max,min} = \sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tau_{max,min} = \pm\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$\left(\frac{\gamma_{xy}}{2}\right)^2$

$\left(\frac{\gamma_{xy}}{2}\right)^2$

Transformation of Three-Dimensional Strain

This case may also proceed from the corresponding stress relations by replacing σ by ε & τ by ɣ/2.

$$\varepsilon_{x'} = \varepsilon_x l_1^2 + \varepsilon_y m_1^2 + \varepsilon_z n_1^2 + \gamma_{xy} l_1 m_1 + \gamma_{yz} m_1 n_1 + \gamma_{xz} l_1 n_1$$

$$\gamma_{x'y'} = 2\left(\varepsilon_x l_1 l_2 + \varepsilon_y m_1 m_2 + \varepsilon_z n_1 n_2\right) + \gamma_{xy}\left(l_1 m_2 + m_1 l_2\right)$$
$$+ \gamma_{yz}\left(m_1 n_2 + n_1 m_2\right) + \gamma_{xz}\left(n_1 l_2 + l_1 n_2\right)$$

⇒ 3-D transformed strains

Similar expressions are found for $\varepsilon_{y'}$, $\varepsilon_{z'}$, $\gamma_{y'z'}$ & $\gamma_{x'z'}$.

ε_x ε_y ε_z $\frac{\gamma_{xy}}{2}$ $\frac{\gamma_{yz}}{2}$ $\frac{\gamma_{zx}}{2}$

$$\boxed{\begin{aligned} \sigma_{x'} &= \sigma_x l_1^2 + \sigma_y m_1^2 + \sigma_z n_1^2 + 2(\tau_{xy} l_1 m_1 + \tau_{yz} m_1 n_1 + \tau_{zx} l_1 n_1) \\ \tau_{x'y'} &= \sigma_x l_1 l_2 + \sigma_y m_1 m_2 + \sigma_z n_1 n_2 + \tau_{xy}(l_1 m_2 + m_1 l_2) + \\ &\quad + \tau_{yz}(m_1 n_2 + n_1 m_2) + \tau_{xz}(n_1 l_2 + l_1 n_2) \end{aligned}}$$

$\frac{\gamma_{x'y'}}{2}$ $\frac{\gamma_{xy}}{2}$ $\frac{\gamma_{yz}}{2}$ $\frac{\gamma_{zx}}{2}$

▼ The principal strains in three dimensions are the roots of the following cubic equation:

$$\varepsilon_p^3 - J_1\varepsilon_p^2 + J_2\varepsilon_p - J_3 = 0$$

Cubic eqn. for ε_p!!

↳ ε_1, ε_2 & ε_3

where

$$J_1 = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

$$J_2 = \varepsilon_x\varepsilon_y + \varepsilon_x\varepsilon_z + \varepsilon_y\varepsilon_z - \frac{1}{4}\left(\gamma_{xy}^2 + \gamma_{yz}^2 + \gamma_{xz}^2\right)$$

$$J_3 = \begin{vmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{xy} & \varepsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{xz} & \frac{1}{2}\gamma_{yz} & \varepsilon_z \end{vmatrix}$$

$$\sigma_p^3 - I_1\sigma_P^2 + I_2\sigma_P - I_3 = 0$$

$$I_1 = \sigma_x + \sigma_y + \sigma_z$$

$$I_2 = \sigma_x\sigma_y + \sigma_x\sigma_z + \sigma_y\sigma_z - \tau_{xy}^2 - \tau_{yz}^2 - \tau_{xz}^2$$

$$I_3 = \begin{vmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{vmatrix}$$

$\left(\frac{\gamma_{xy}}{2}\right)^2$ $\left(\frac{\gamma_{yz}}{2}\right)^2$ $\left(\frac{\gamma_{zx}}{2}\right)^2$

Mohr's Circle for strain

-> Read page 77 "Mohr's Circle for strain"